

**A NEW VISION OF TEAM KNOWLEDGE:
MULTIDIMENSIONAL CONTENT & STRUCTURE**

Mark A. Clark
Kogod School of Business, American University
4400 Massachusetts Avenue NW
Washington, DC 20016-8044
(202) 885-1873
Fax: (202) 885-2691
mark.clark@american.edu

J. Alberto Espinosa
Kogod School of Business, American University
4400 Massachusetts Avenue NW
Washington, DC 20016-8044
(202) 885-1858
alberto@american.edu

A NEW VISION OF TEAM KNOWLEDGE: MULTIDIMENSIONAL CONTENT & STRUCTURE

ABSTRACT

In spite of increasing interest in team knowledge, as evidenced by the multiple perspectives discussed in the literature, the lack of an integrated framework limits both our understanding of its boundaries and the efficacy of its measures. We propose that team knowledge may be conceptualized as sets of dimensions related to (1) the content of knowledge; and (2) how this knowledge is held by team members – that is, its structure. This paper builds on a range of theoretical perspectives to propose a multidimensional construct of team knowledge in which the various dimensions are organized as pertaining to either *content* or *structure*. We further discuss measurement issues related to each dimension, detailing empirical methods previously used and suggesting further approaches. Finally, we illustrate how these multiple perspectives on team knowledge can be explicated through combinations of our suggested dimensions.

Keywords: team knowledge, team mental models, knowledge dimensions

A NEW VISION OF TEAM KNOWLEDGE: MULTIDIMENSIONAL CONTENT & STRUCTURE

Understanding the manner in which work teams hold and use knowledge has become important as organizations increasingly depend on teams to process information and accomplish a variety of tasks (Anand, Clark, & Zellmer-Bruhn, 2003; Cohen & Bailey, 1997). Although many studies have focused on team knowledge, it has not been comprehensively defined, instead borrowing elements from relatively macro knowledge management (Nonaka, 1994), micro cognitive science (Adams & Pew, 1995), and process-oriented information theories (Hinsz, Tindale, & Vollrath, 1997). To date, our understanding of team knowledge has advanced primarily through examining a single perspective at any one time, such as when borrowing Polyani's (1966) explicit-tacit taxonomy, and through various calculations of team mental models (e.g., Cramton, 2001; Klimoski et al., 1994). There have been few attempts to consider team knowledge as a multidimensional construct (Anand et al., 2003 is one exception), and none that attempt to reconcile the various measures used. This lack of an integrated framework limits both our understanding of the boundaries of team knowledge and the efficacy of team knowledge measures.

Knowledge can be viewed as a state of mind acquired from experience or study (Alavi & Leidner, 2001) and it can be individual or collective (Nonaka 1994). Therefore, team knowledge is composed of the individual and collective aspects of the various states of mind possessed by all members of the team. But team knowledge is more complex than the simple aggregation of the individual knowledge of all team members. A team member will have different patterns of interaction and interrelation with other team members depending, not only on the content of knowledge possessed by that team member, but also on what content of knowledge is possessed by other team members and how this structured across team members. Therefore, we argue that

in order to develop a clear conceptualization of team knowledge, researchers need to take into account two different but interrelated aspects of knowledge: (1) *content* – the pool of knowledge available in the team; and (2) *structure* – how the knowledge is held and organized across team members. This paper builds on a range of perspectives to propose a multidimensional construct of team knowledge, organized as dimensions pertaining to the *content* or *structure* of team knowledge. This framework seeks to detail aspects of team knowledge most relevant to team performance, rather than providing an exhaustive set of descriptors (including, for instance, aspects more relevant to individual well-being). Importantly, our discussion includes an operational perspective, detailing empirical methods that have been used to measure each dimension and suggest further approaches. Finally, we illustrate multiple perspectives on team knowledge through combinations of our suggested dimensions.

DIMENSIONS OF WORK TEAM KNOWLEDGE: CONTENT & STRUCTURE

Knowledge, whether held in organizations, teams, or individuals, has been a popular construct for both academics and practitioners; however, our understanding of knowledge as a construct is incomplete (Hammer, Leonard, & Davenport, 2004). Some have described dimensions of knowledge that relate to its content, as when Schultz (2001) defined knowledge as a broad concept “learned from experience or study... includ[ing] insights, interpretation, and information” (p. 662). Knowledge content can vary, for instance, in its level of maturity (Berman, Down, & Hill, 2002) or accuracy (Espinosa, 2001). Others have pointed out that knowledge varies in its structure, its readiness “to apply to decisions and actions” (Davenport, De Long, & Beers, 1998, p. 43); for example, whether individuals hold knowledge internally or simply have access to external sources (Anand, Manz, & Glick, 1998). As can be seen, knowledge is a complex construct that cannot be wholly described by a single dimension. This

complexity is exacerbated at the team level, where structural considerations grow beyond individuals, and content includes knowledge about both task and team characteristics.

For example, members may have complementary knowledge or their knowledge may be redundant. Their knowledge may also vary in terms of accuracy. Some teams may have evenly distributed knowledge, while other team members may have highly differentiated and specialized knowledge. All these different possibilities will give rise to complex team knowledge structures that cannot be explained by knowledge content alone. We therefore suggest that team knowledge may be best articulated as a multidimensional construct, with individual dimensions characterizing relative degrees of knowledge content and structure. To this end, we offer a set of dimensions related to team knowledge content (relevant to a particular domain; pertaining to the applicability to task or outcome) – *task- vs. team- oriented, maturity, accuracy, and time span* – and a set related to team knowledge structure (how the knowledge is held, processed, or formatted to be usable) – *tacit or explicit, sharedness, externalization, and differentiation*. We posit the dimensions to be relatively independent and complementary, if not completely orthogonal. Further, we believe that this set of dimensions is most related to the operation of real-world workteams, although we recognize that inclusion of further dimensions is an arguable point. Indeed, we hope that this effort will stimulate additional conceptualization of the team knowledge construct.

In the following sections, we describe the theoretical foundations for these dimensions and raise important related measurement issues.

DIMENSIONS OF TEAM KNOWLEDGE CONTENT

The first set of dimensions describes the content of team knowledge; the characteristics pertaining to the task, outcome of interest, or any particular domain related to team functioning.

Dimension 1: Taskwork vs. Teamwork

Theoretical foundations. Team cognition researchers suggest that most of the knowledge that really matters for working teams can be classified in one of two categories: taskwork knowledge – the knowledge required to carry out the task itself; and teamwork knowledge – the knowledge required to interact and work with other team members (Cooke et al., 2000; Klimoski et al., 1994; Rentsch et al., 1994). This effectively classifies the various team knowledge constructs discussed in the literature into one or the other category. For example, task awareness is up-to-the minute knowledge of what is happening in the performance context (Endsley, 1995) – a type of taskwork knowledge. Similarly, the concept of transactive memory is the knowledge of what others in the team know (Wegner, 1986) – a type of teamwork knowledge.

It is important to note that both taskwork knowledge and teamwork knowledge can encompass many domains. Taskwork knowledge includes knowledge about technical concepts, products, and processes as well as the overall context in which the task is taking place, which is important for coordination and performance. For example, studies of software teams have found that the leading factor associated with mistakes in software development is the thin spread of application domain knowledge within the team (Curtis, Krasner, & Iscoe, 1988) and that taskwork knowledge sharing and integration is essential for performance (Walz, Elam, & Curtis, 1993). Similarly, teamwork knowledge includes knowledge about what team members know, where they are, and what processes they use to interact, which helps teams to interact more effectively and to achieve high levels of coordination. For instance, studies of large-scale software teams have found that knowing where expertise resides in a team and when it should be accessed had a positive effect on team performance (Faraj et al., 2000). Further, team members

who understand of how their actions affect others' tasks – a collective mind of sorts (Weick & Roberts, 1993) – were better coordinated with one another (Crowston & Kammerer, 1998).

Measurement issues. Consistent with the recommendations of conceptual and empirical work in the team cognition literature (Cooke et al., 2003; Espinosa & Carley, 2001; Levesque et al., 2001; Mathieu et al., 2000), we argue that measures for the taskwork vs. teamwork knowledge dimension must be based on measures of each member's knowledge of relevant areas and on aggregation and distribution measures. Also, in the case of taskwork knowledge, it is important to measure the knowledge that each member has about each key task area or related concepts, products, technologies or processes. In the case of teamwork knowledge, the knowledge that each member has about each of the other members regarding each key aspect of teamwork activities and processes should be measured.

Dimension 2: Maturity (Emergent vs. Mature)

Theoretical foundations. Team knowledge content will also vary in its level of maturity (cf. Berman et al., 2002), characterized by its level of readiness for application to the performance objective (dynamic properties comprise a separate dimension, discussed below). Knowledge emerges as a product of the interaction between team members, although in such shared contexts emergent knowledge is not necessarily newly created. For instance, knowledge that is held only by one or more members may be transferred to others through exchange of ideas, information, or procedures, thus emerging in the group context. Other knowledge may represent novel combinations of previous information or new insights, thus emerging as created knowledge. In either case, this emergent knowledge may then be subject to influence from a variety of task and team factors and processes as it matures into knowledge that the group can agree upon. More mature knowledge solidifies into a cognitive form that has a certain amount of

stability or institutional inertia, becoming a norm, plan, or a way of completing tasks.

Emerging knowledge may arise as a perspective or line of inquiry held by one or more team members, or as a combination of differing perspectives in the case of created knowledge. As it emerges, knowledge may be shaped by group process factors such as power dynamics and interpersonal diversity. The locus of power in a team may influence which aspects of knowledge will be accepted by the group, while diversity of backgrounds, values, and personality may thwart communication efforts (Zenger & Lawrence, 1989) and possibly create factions (Lau & Murnighan, 1998) which may impede knowledge transmission or influence its interpretation (Clark, Anand, & Roberson, 2000). More task-related aspects of the knowledge such as accuracy and functionality may also direct the development of emerging knowledge.

More mature knowledge will be less susceptible to influence from such dynamics and will generally have increased resistance to change or revision, sometimes to the point of “ossification” (Berman et al., 2002). Team members will be more familiar with both the explicit and tacit aspects of the knowledge, accepting or at least abiding by its character. Mature team-level knowledge is likely to be present in a team that has had significant time or experience together. At those later points in the lifecycle, members’ stores of knowledge are likely to have been exposed to other team members, reducing the overall differentiation of the team (Gersick & Hackman, 1990). At this point, team members may also have less need to communicate when utilizing knowledge or engaging in routine procedures.

Measurement issues. In line with the conceptual issues discussed, the measurement of team knowledge should vary according to its level of maturity. Emerging team knowledge in the form of innovation has been measured as aggregate performance on simulated tasks, such as a flight software package (Ellis et al., 2003). As knowledge creation, it has been measured through

network analysis, determining the number and strength of associations of co-authors through journal citations (McFadyyn & Cannella, in press). In emerging stages, it will be important to account for the effect of group process factors (as moderators and intervening variables) on knowledge structure, transmission, and use.

It is important to note that the temporal emergence of knowledge is to also have an influence on knowledge differentiation in teams. As knowledge emerges through transfer or creation, it is likely to be more differentiated than more mature knowledge that has survived the dialogue and negotiation process which renders it more palatable to a general interpretation (Anand et al., 2003). Related to this process, it may be that teams with more mature knowledge – which has likely been through an interpretation process – may not need much interaction on task-focused issues (cf. Clark et al., 2000). When measuring knowledge, especially when doing so through network analyses, it may be important to allow for the affects of maturity on differentiation (separate domains of knowledge) and the extent to which team members are conferring or interacting with one another.

Dimension 3: Accuracy

Theoretical foundations. While team knowledge is very important for coordinating task activity, inaccurate knowledge may seriously thwart team objectives and performance. Accuracy can be defined as propinquity to correct or desired values. Many studies have assessed team knowledge through surveys in which participants are asked to relate different aspects of a task (Mathieu et al., 2000; Rentsch et al., 2001) or to ask task or team related questions (Espinosa et al., 2002a; Levesque et al., 2001) and then computed similarity statistics to measure shared knowledge. Yet other studies have asked general questions about the team in general (e.g., “I know which team members have expertise in specific areas”, Lewis, 2003). While these items

are useful to measure different aspects of team knowledge, they don't control for the accuracy of the responses. For example, the measures that compute knowledge similarity only measure whether the knowledge of two team members converge, but the knowledge of these two members may be incorrect knowledge, which is particularly the case when teams fall into situations of groupthink or in cases where teams have operated together for a long time without access to external knowledge to help them validate their internal knowledge.

Measurement issues. The measures along the different dimensions of team knowledge described in this paper need to be corrected or controlled for the accuracy of that knowledge. Team knowledge accuracy can be measured by comparing member's shared knowledge with that of a referent domain expert (Cooke et al., 2000) or some other referent source (Cooke et al., 2003). When a domain expert is not available, accuracy can be measured by either comparing members' knowledge to actual facts or by having members rate each other on different aspects of their task knowledge. For example, in the case of taskwork, if several team members are rating several knowledge items in a questionnaire and no referent knowledge source is available, one can evaluate what the team consensus is on these items and compute deviations of individual responses from this consensus (Espinosa, 2001). While this does not provide a true measure of accuracy and does not correct for team biases, it is likely better than having no measure of accuracy at all. Similarly, in the case of teamwork, if no referent knowledge source is available, individual ratings of other members' knowledge can be compared to team consensus about that member or to self-rated measures of knowledge. Again, this does not correct for team biases. Of course, in both cases, it is best to have a good source of referent knowledge to benchmark team members' knowledge for accuracy.

Dimension 4: Time Span (Long-lasting vs. Fleeting)

Theoretical foundations. Time span refers to how long-lasting the relevance of the particular knowledge is for the task (rather than how long ago was the knowledge acquired). Most of the team cognition literature deals with either long-lasting or fleeting knowledge, but it rarely deals with both despite their distinct natures (Cooke et al., 2003). Long-lasting knowledge is acquired through experience with a task domain, and by interacting and training with other team members and it is relevance to the task over long periods of time or even the entire duration of the task (e.g., knowledge of a software programming language). Representations of this include team mental models (e.g., Carley, 1997; Kraiger et al., 1997; Mathieu et al., 2000), schema similarity (Rentsch & Hall, 1994; Rentsch & Klimoski, 2001), knowledge similarity (Espinosa et al., 2002a); shared knowledge (Nelson et al., 1996), and transactive memory (Wegner, 1986).

Team members also need more immediate fleeting knowledge, relevant for specific task situations. One important property of awareness knowledge is that its relevance diminishes when the situation ends. For example, a pilot flying an airplane needs to be attentive to instrument readings and weather conditions during the flight, but once on the ground such information is no longer relevant and can therefore be forgotten. Most discussions of fleeting knowledge in the literature derive from “situation awareness”, a concept that has been studied intensively in military and aviation contexts, defined as “up-to-the minute” knowledge necessary to interact with a human or physical system (Adams, Tenney, & Pew, 1995). Situation awareness in a team context can be defined as understanding how events in the task environment, such as the activities of other team members, affects the future of the task and one’s own activities (cf. Dourish & Bly, 1992; Endsley, 1995). Thus, team members should not only have situational awareness to carry out their task responsibilities, but also team situation awareness to work

effectively as a coordinated unit (Wellens, 1993).

It also should be noted that the concept of situation awareness has been extrapolated to many aspects of team collaboration, including task awareness (Espinosa et al., 2000), workspace awareness (Gutwin & Greenberg, 1999; Gutwin & Greenberg, 2004); presence awareness (Boyer, Handel, & Herbsleb, 1998), and others (Steinfeld, Jang, & Pfaff, 1999). Our terms, long-lasting knowledge and fleeting knowledge, are similar to the categorization proposed by Cooke and her colleagues (2000), but more inclusive of the various types of knowledge content which may be represented.

Measurement issues. Consistent with the recommendations provided by Cooke (Cooke et al., 2003), we argue that measures for long-lasting team knowledge versus fleeting team knowledge must include several considerations. First, measures of long-lasting team knowledge must be based on accurate measures of each member's knowledge in different task-relevant areas and on aggregation and distribution measures. Because we are interested in measuring long-lasting knowledge, the knowledge elicitation methods should be based on items that diagnose the actual knowledge of team members. Therefore, recall-based instruments like surveys and interviews are appropriate to measure this type of knowledge. Archival data based on knowledge repositories like documents and text files are also appropriate. Second, measures based on recall of actual knowledge are not appropriate for team awareness measures. Because team awareness knowledge changes relevance quickly as the situation changes, participants are not likely to recall much awareness knowledge they possessed for a given past situation. Therefore, it is more appropriate to have event driven instruments in which participants' team awareness is evaluated as the specific situations are occurring. This can be done with multiple short surveys administered at random times concurrently with the situations being measured.

DIMENSIONS OF TEAM KNOWLEDGE STRUCTURE

The second set of dimensions describes the structure of team knowledge: how the knowledge is held, processed, or formatted to be usable to engage in task performance or interact with other team members.

Dimension 5: Tacit vs. Explicit

Theoretical foundations. The content of team knowledge may vary in its tacit or explicit form (Polyani, 1958). Explicit knowledge can be detailed and communicated without much difficulty; for instance, a written manual for an organization's recruitment process outlines explicit procedures. Tacit knowledge is not so easily articulated; it is procedural and "sticky", and as such is more difficult and time consuming to transfer between individuals (Kogut & Zander, 1997). Examples of tacit knowledge include skill in servicing complex machines and abiding by idiosyncratic norms of interpersonal interaction (Nonaka, 1994). A simple division between tacit and explicit, however, belies what is often a strong association between the two knowledge forms. A recruitment manual will lay out organizational standards, for example, but there will most often be a great deal of less explicit, but related, knowledge that is necessary to actually complete the job. Learning the tacit aspects of a task most often requires some increased level of interaction between organizational members, ranging from simple observation to coaching and actively working together on tasks.

Due to its inscrutable nature, tacit knowledge is more susceptible to being lost or incorrectly translated from member to member. It is especially vulnerable to the effects of turnover at the team or organizational level, or poor training and mentoring programs. When properly supported within a team or organizational environment, tacit knowledge may be less susceptible to decay than simple written or spoken instructions (Cohen, 1996). It also may be

more difficult to track transmission of tacit knowledge. Organizations which have close relationships with suppliers and clients may have to create multi-organizational task forces to promote knowledge transfer, a resource-intensive solution which also risks knowledge hemorrhage to outside parties.

Measurement issues. By its nature, explicit knowledge is the easier of the two types to measure. Fairly direct approaches can be used to assess content, process, or outcome. Content measures include technical or task-oriented questions in relatively lean media (e.g., written). Process measures include dialogue counts, through direct observation, groupware, or other means. Outcome measures involve goal achievement assessment such as reports, patents, or other output. Cognitive mapping techniques have been used to represent explicit knowledge at individual (Axelrod, 1976), team (Carley, 1997), and organizational levels (Weick & Bougon, 1986) by charting “meaningful patterns of ... concepts and connections between them” (Kuhn & Corman, 2003, p. 206). Such mapping may also reveal tacit themes that link the shared concepts. Several measures have been used to ascertain cognitive maps, including repertory grids (Kelly, 1955), Q-sorts (Daniels, Johnson, & Chernatony, 1994), textual analysis of open-ended questions, (Carley, 1997), coding interviews into predefined categories (Carley, 1986), and centering resonance analysis (CRA), which assesses conceptual centrality through adjacency matrices (Corman et al., 2002). The CRA technique is paired with multi-dimensional scaling to compare schema and draw a map of collective mental models.

Measures of tacit knowledge tend to be more indirect. It might be possible to ascertain shared tacit knowledge through some form of network analysis (Droege & Hoobler, 2003), compiling instances of people working together, such as in mentoring relationships, on-the-job training, or shared time in work teams (as in professional basketball squads, as noted by Berman,

Down, & Hill, 2002). Other work suggests that tacit knowledge may be transmitted, and presumably measured, through thematic storytelling, in which themes are evoked to teach a lesson convey or a norm (Boje, 1991). Individual tacit knowledge might be assessed through successful completion of activities requiring such knowledge, such as driving a bulldozer, executing a gymnastic maneuver, finishing a complex task, or the process of joining – integrating one’s skills, knowledge, personality, etc., into – a team. Relatively rich communication media, such as topical chat room discussions (rather than email) may indicate tacit knowledge transfer.

Dimension 6: Shared vs. Individual

Theoretical foundations. While it is accepted that knowledge can be held individually or collectively (Nonaka, 1994), team knowledge has typically been characterized as knowledge shared between members. We believe, however, that team knowledge should be thought of as the combination of both components: the individual knowledge possessed by each team member that is not possessed by other members, as well as the knowledge that is shared among members (note that the shared knowledge component includes some individual knowledge; therefore we refer to shared and unshared, rather than shared and individual). While both types of knowledge are important, individual unshared knowledge can be most beneficial to the team when others have some understanding of what this knowledge is and are able to access it when need it (cf. Faraj & Sproull, 2000).

While individual knowledge is very important for performance in individual tasks, the team cognition literature suggests that shared knowledge is important to coordinate the activities in these individual tasks so that the respective dependencies among these activities are effectively handled. The individual unshared knowledge possessed by experts, combined with the shared knowledge that exists among members, creates holistic team knowledge structures

that are different and more complex than the aggregation of the parts (Cooke et al., 2003).

Shared knowledge provides a common knowledge base through which team interaction can occur, which enables team members to tap into expert knowledge sources in the team (Alavi et al., 2001). It helps members synchronize their actions based on accurate expectations about what others in the team are likely to do (Wittenbaum & Stasser, 1996).

The shared portion of team knowledge has been characterized in a number of ways, including: shared mental models – i.e., organized knowledge that team members share about the task and each other, acquired by working, interacting and training together, which helps members form accurate explanations and expectations about task activities (Cannon-Bowers, Salas, & Converse, 1993; Klimoski et al., 1994; Kraiger & Wenzel, 1997); mutual knowledge – i.e., knowledge that team members share and know that they share it, which improves the common ground among team members, making their communication more effective (Clark & Carlson, 1982; Clark & Marshal, 1981; Cramton, 2001; Fussell & Krauss, 1992; Krauss & Fussell, 1990); shared knowledge – that is, an understanding among team members for the technologies and processes that affect their mutual performance (Nelson & Coopridier, 1996), which provides members with the ability to access knowledge sources in the team and apply them effectively when necessary (Alavi et al., 2001); and transactive memory – knowledge of who knows what in the team – which helps teams determine who to contact when they to access others' individual knowledge when needed and assign task responsibilities to the respective specialists in the team (Wegner, 1986, 1995).

Measurement issues. There are a number of measures that have been proposed in the research literature to assess shared team knowledge (Carley, 1997; Cooke et al., 2003; Espinosa & Carley, 2001; Espinosa et al., 2002b; Levesque, Wilson, & Wholey, 2001; Mathieu, Goodwin,

Heffner, Salas, & Cannon-Bowers, 2000; Rentsch & Klimoski, 2001). While these methods have subtle differences, they share three attributes. First, all methods measure some form of knowledge similarity in the team. Some methods look at similarity of how individuals interrelate different parts of the task and then compute a correlation or other similarity score among these different individual mental schemas (Cooke et al., 2003; Mathieu et al., 2000; Rentsch et al., 2001). Other methods look at similarity of knowledge content among team members (Espinosa et al., 2002b; Levesque et al., 2001). And yet another method looks at the similarity of words and sentences used in communication among team members (Carley, 1997). Second, except for one study (Espinosa et al., 2002a), none of these methods account for individual knowledge, therefore they do not provide a holistic measure of team knowledge. And finally, except for one study (Espinosa et al., 2002a), these methods provide aggregate measures of shared knowledge for the entire team, but do not account for how this knowledge is distributed in the team.

Consistent with methods proposed in prior work (Espinosa et al., 2001), we argue that measures for the shared vs. unshared dimension of team knowledge must include a number of things. First, they must include measures of the knowledge of each individual team member in each key task-relevant area. These can be obtained from scales that ask a number of task relevant questions for which the correct answer is known. Many measures used rely on survey scales about perception of knowledge (e.g., “I know my team members well), which may be inaccurate. Second, measures should also assess the knowledge similarity and knowledge distance between every pair of members in the team. This is important because knowledge can be shared between any two individuals in the team, regardless of whether all members share it. An aggregate of this can be computed using correlation scores, within-team agreement scores such as R_{WG} (James, Demaree, & Wolf, 1984) or any other similarity metric. The dyadic knowledge similarity metrics

can then be aggregated to compute an aggregate shared knowledge metric. Third, these measures must also include some measure of knowledge specialization or knowledge dissimilarity among team members, which can be computed using a distance metric (e.g., inverse correlation, absolute value of differences). Finally, these measures should also assess knowledge distribution within the team, capturing distribution across both task knowledge areas and members. Network analysis methods provide excellent tools to describe important properties of networks like centrality, isolation, and structure, which can make a difference in knowledge networks.

Dimension 7: Internal vs. External

Theoretical foundations. In order to accomplish their task directives, teams may choose to make use of external knowledge sources in addition to the knowledge held by the members (internal to the team). Members depending on external resources do so through the use of some form of directory structure, rather than by acquiring the knowledge for themselves (Anand, Manz, & Glick, 1998). A directory structure can be understood as a mental telephone book, listing knowledge resources outside the team, similarly to the “knows who knows what” transactive memory model that has been used to describe the internal knowledge structure (Wegner, 1986). Utilizing externalized knowledge means that members may never personally possess the knowledge. Rather, they use a variety of techniques that allow external knowledge to be used in task completion. For instance, a team may delegate a task to a non-member, or it may invite outsiders to specific team meetings to work on designated aspects of the team’s task. Teams differ in the amount of external knowledge that they use because they may develop different norms about knowledge sources. Some teams value the use of external knowledge, while other teams mistrust external knowledge and do not use it even when it would be advantageous. Additionally, the team’s task may preclude use of external knowledge. For

instance, a personnel committee evaluating performance of key executives may be unable to use external knowledge due to confidentiality requirements.

Internal knowledge is important to the functioning of a team. The translation of knowledge from both explicit to tacit and from tacit to explicit is accomplished through the team; diffusion of both forms leads to its acquisition, integration, and storage by the various team members (cf. Boland et al., 2001).

Measurement issues. External knowledge used by a team may be measured through network analysis and related means. Team members' external contacts may be counted through social or expert networks (Leibiskind, Oliver, Zucker, & Brewe, 1996), or member reports of which external sources have been utilized (Tsai, 2002). Weak ties between project teams have been linked to speed of product development (Hansen, 1999). Intergroup social networks have also been measured for their role in thwarting team performance (Sparrowe, Liden, Wayne, & Kraimer, 2001). An alternate measure would be the number and type of external consultants (both internal and external to the organization) or outsourced work employed by a team to complete its task. A further, less direct method would be to determine whether teams used knowledge outside their known areas of expertise.

Internal knowledge use may be similarly measured. It might first be necessary to codify a knowledge map or transactive memory structure such that internal knowledge held by the team is apparent. Then network density measures may be used to determine the extent of connection (similarity in knowledge), and possibly utilization (knowledge usage), between members (Reagans & Zuckerman, 2001). Others have conceptualized internal team knowledge as awareness of one another's knowledge (Austin, 2003).

Dimension 8: Differentiation – Even vs. Distributed

Theoretical foundations. The knowledge held by a team can also be considered in terms of its distribution among the members: whether the members know similar or different things (Wegner, 1986), and how evenly this knowledge is dispersed. Members of a highly differentiated team possess knowledge in separate domains, creating differences not only in knowledge base but also in world-view and motivation (Lawrence & Lorsch, 1967). Cross-functional product development teams, for instance, can be expected to have differences not only in the specialized knowledge that each member brings to the team, but also in the value that members place on end goals (e.g., utility versus marketability) and the procedures used to complete the task.

The dispersion of knowledge can also vary greatly in a team. A highly differentiated team may have an even distribution of knowledge, such that each member has expertise in a separate field. Alternately, differentiation could be the result of relatively few members having very distinct knowledge stores, or distribution could occur through factions of the team sharing similar knowledge that other members or factions do not share. The development of such factions decreases the likelihood of knowledge-sharing across functional lines, as an in-group – out-group dynamic is introduced (Tajfel & Turner, 1985). In addition to dispersion of a single knowledge domain across members, multiple domains may combine to form “faultlines” (cf. Lau & Murnighan, 1998), which impede the ability of the resultant factions to understand one another beyond simple additive effects of each difference.

Measurement issues. As discussed above, several measures have been used to ascertain the extent to which knowledge is shared within a single domain. To measure differentiation, some decision must be made as to the boundaries of the domain of interest. For example, a team of automotive designers could be said to have low differentiation if the domain is defined as auto design, but high differentiation if the domain was defined as knowledge of particular subtasks

such as molding fixtures or carpeting contours. Practically, this means that measures comprised of a given set of scales must be conceptually validated in association with the given task and outcomes of interest before differentiation can be determined.

In controlled research designs, some team knowledge measures have been empirically validated as relatively orthogonal (Cooke et al., 2003; Stasser et al., 1995). This rather nuanced approach has not typically been accomplished in field research, where knowledge differentiation has often been more broadly defined in terms of functional background (Bantel, 1993; Pelled, Eisenhardt, & Xin, 1999; Smith, 1994) and education (Jackson et al., 1991; Murray, 1989).

In field studies, the distribution of different types of knowledge within a team has generally been assessed with measures of central tendency (typically mean) and variance (typically standard deviation). This measure is not concerned with the content of the knowledge domain, but whether members of the team have similar stores or experiences. Mapping this distribution further is the measure described by Espinosa, Carley, and their colleagues (2001), which attempts to account for how knowledge about different task domains is distributed within a team. A next step may involve computing some diversity metric using a Herfindahl index, and entropy measure, a Gini coefficient, or any other diversity metric.

DIMENSION MATRICES: ILLUSTRATING KNOWLEDGE FORMS

Up to this point we have attempted to isolate the discussions of dimensions, describing and illustrating simplified “types” knowledge to enable exploration of the limits and possibilities of the construct (Doty & Glick, 1994). However, we recognize that such partitions don’t often occur in real organizations, and that these dimensions may be best used jointly to describe many team knowledge instances. We offer a few examples below that pair the extremes of two dimensions – for instance, long-term and dynamic knowledge paired with individual and shared

forms of knowledge. Our account of the resultant 2x2 matrices includes probing the influence of various factors (e.g., accuracy) on each pairing where appropriate. Grouped together, key dimensions can help us to categorize and understand how members hold and use knowledge within their workteams.

Illustration of the Sharedness and Time Span Dimensions

Table 1 presents a knowledge typology illustration from the perspective of two of the dimensions we discussed: (1) individual vs. shared; and (2) long-term vs. dynamic. We discuss the resulting types in each cell.

 Insert Table 1 about here

Individual, long-term knowledge. This knowledge, acquired through education, training and experience with the task domain, enables members to perform their individual tasks. Individual task knowledge can be very homogeneous or highly differentiated, regardless an individual's knowledge about other tasks or non-task domains. For example, a code inspector in a software team needs to know the specific software language used to program the code and the application domain (Curtis et al., 1988). In contrast, a senior manager in a strategic planning team will need to apply knowledge from a wide range of domains (e.g., finance, marketing, production, competition, legal environment, industry, etc.).

Shared long-term knowledge. This is the knowledge similarity or overlap that members have on the content or structure of their individual knowledge, such as that conceptualized through shared mental models (Cannon-Bowers et al., 1993; Klimoski et al., 1994). This type of shared knowledge develops slowly over time through training and experience working with

others in the task and is long-lasting (Cooke et al., 2000), and it is particularly important for communication and coordination.

Individual, Fleeting: Individual situation awareness. Individual situation awareness involves perceptions about what is happening in the task environment, comprehension of the implications of the situation, and the mental projection of how the situation will affect future states of the task environment (Endsley, 1995). For example, a software testing engineer needs to know about the availability to testing lab facilities, and a senior management preparing a strategic plan needs to know if a particular supplier has gone on strike.

Shared, Fleeting: Team awareness. As with long-term knowledge, individual situation awareness is important for individual performance while team awareness is important for team coordination (Endsley, 1995; Wellens, 1993). This may involve developing awareness of task activities, the presence of others, or how the shared workspace has changed, among many other things. For example, a software testing engineer needs to be aware if the development of a piece of software code is finished and ready for testing, or if it is behind schedule and by how much.

Illustration of the Taskwork/Teamwork and Time Span Dimensions

Table 2 presents another typological illustration from the perspective of two other knowledge dimensions we discussed: (1) taskwork vs. teamwork knowledge; and (2) long-term vs. dynamic knowledge.

 Insert Table 2 about here

Long-term knowledge of taskwork. Taskwork involves all the activities that need to be carried out by team members to carry out the task (Klimoski and Mohammed 1994). Long-term

knowledge of taskwork includes all the task knowledge that team members brought from their prior experience in the task domain, plus all the knowledge developed and acquired during the task, which are necessary to carry out task activities. This knowledge develops slowly in the form of mental models or schemas, which involves how different parts of the task relate to each other, and taskwork knowledge content (Cannon-Bowers and Salas 2001), and it is long-lasting.

Long-term knowledge of teamwork. Teamwork involves all the activities that need for team members to interact and work with each other (Klimoski and Mohammed 1994). Long-term knowledge of teamwork includes all the prior knowledge that team members had about each other before the task started, plus all the knowledge developed and acquired about each other during the task. This knowledge also develops slowly as team members interact over time and is long-lasting. The individual knowledge that team members have about others in the team has been referred to in the literature as “transactive memory” (Wegner 1995), which helps members find sources of knowledge when needed (Faraj and Sproull 2000) and develop expectations of who is supposed to pay attention to which type of new knowledge when it arrives, depending on who has which expertise. But knowledge of teamwork also involves knowledge about team processes, interaction patterns and routines.

Team awareness of taskwork. Team members need to have individual situation awareness of taskwork to carry out their individual task activities and team awareness of taskwork to understand what is happening with the activities of others, which may affect their own task activities (Wellens 1993; Endsley 1995). This knowledge is fleeting in the sense that it is acquired quickly, it is only relevant to a particular situation, and its relevance goes away when the situation passes. Awareness of taskwork is important for members to understand how the task environment is changing and how this change will affect their own task activities.

Team awareness of teamwork. As with taskwork, team members need to have individual situation awareness of what their team members are doing and how their actions are affecting their own task activities (Wellens 1993; Endsley 1995). Similarly, this knowledge is fleeting in the sense that it is only relevant when the particular actions of the team members involved can affect one's task activities, and the need for this team member awareness when the actions of that member are ceased to be relevant to the immediate task. Awareness of team members can be very specific about one member or one type of member awareness, or it can be more general about several members or several types of awareness (e.g., presence awareness, chronological awareness).

Illustration of the Taskwork/Teamwork and Explicit/Tacit Dimensions

A further combination of knowledge dimensions is illustrated in Table 3, pairing the explicit-tacit dimension with the content – knowledge about the team versus knowledge about the task.

Insert Table 3 about here

Declarative task knowledge. Teams engaged in conjunctive tasks have a set amount of task-related, explicit knowledge that is held by individual members and in archival storage (e.g., manuals). The members of such teams also have varying awareness of the content of that declarative knowledge, where it is stored (within or outside of the team), how it is distributed, and how accurate it is. Within these constraints, declarative task knowledge is probably the most straightforward of the four pairs to measure: it is declarative knowledge about methods to complete familiar or repeated tasks, typically characterized by mutual agreement on the meaning

of concepts and the definition of terms. It has been measured through textual analysis (Carley, 1997; Mathieu et al., 2000), shared training (Stasser et al., 1995), functional background – various top management team researchers, including Bantel (1993), and other team researchers such as Pelled, Eisenhardt, and Xin, (1999) – and shared experience on a work team (Espinosa et al., 2002b).

The knowledge that will actually be used in a collective task context is not always straightforward. Libby, Trotman, and Zimmer (1997) found that differential expertise must be recognized to be useful; Baumann and Bonner (2004) found that members whose task expertise was recognized by other members were deferred to less than two-thirds of the time. This implies that team members often don't value accuracy as highly as they perhaps should.

Procedural knowledge/ability. Tacit knowledge by its nature is more difficult to measure. Such procedural understanding underlies the conversations present in a team idea exchange, or in the back-and-forth dialogue between members with opposing viewpoints (Clark et al., 2000). As discussed previously, tacit knowledge has a certain strength, enduring once established, vulnerable more to team member turnover than decay over time (cf. Cohen, 1996) . This strength may be evident through measures of task completion and solidified, unquestioned procedures for task achievement.

Team process ability. In contrast to the tacit knowledge used to complete team tasks, process ability refers to the innate know-how required to deal effectively with other team members and engage in task resolution processes. This includes an understanding of team norms concerning these processes and how to tailor these processes for individual members of the team. As such, team process ability is especially susceptible to group dynamics and process issues such as conflict, diversity, and communication. However, the tacit understanding necessary to cope

effectively with team processes implies a fairly high level of development, which may help tacit knowledge to “stick”.

Team process knowledge. Team process knowledge is related to team process ability in that both are necessary to keep the team functioning properly. Team process knowledge, however, will be composed of more codified information such as personnel policies, task deadlines, and formal network directories. This may be evidenced by task completion rates, reduced errors, fit of team process into planned work cycle, and formal personnel policies.

DISCUSSION AND LIMITATIONS

Conceptualizing a team knowledge construct that may be useful is necessarily a complex undertaking. In this paper, we have attempted to outline such construct dimensions that are relevant to measurement and, in turn, empirical relationships with team performance outcomes. We believe this approach to have several advantages. First, it is well grounded in the research literature, incorporating findings from both lab and field research. Second, the use of multiple dimensions enables us to bring together various theoretical perspectives of team knowledge, which may otherwise appear to be dissimilar or incompatible. Third, the dimensions are categorized according to their relevance to the content or the structure of team knowledge. Fourth, by combining any two or more dimensions into a knowledge matrix typology, one is able to represent a number of different descriptions of team knowledge found in the literature. Finally, the proposed construct has clear implications for the development of team knowledge measures. Overall, this approach contributes to our understanding of team knowledge by offering a viable construct that can help to guide team knowledge measure development, which in turn will facilitate much needed empirical research in this area.

We acknowledge several limitations to our approach. First, we have not specifically

discussed all possible outcomes of team knowledge; rather, we have defined the construct in relation to team knowledge structures and their affect on team performance. Next, while well grounded on the research literature on team knowledge, our construct needs empirical validation. Naturally, before conducting related empirical studies, we need to develop measures for the proposed dimensions. Third, we acknowledge that the team knowledge construct may be more complex than we have described in this paper. Therefore, the set of dimensions presented is not exhaustive, but rather illustrative of what we believe are the essential aspects.

For example, we have not discussed a dimension to capture whether the knowledge relates to how the knowledge is organized in people's minds – that is, intrapsychic knowledge structure – or whether it relates to the actual content of this knowledge. Team cognition researchers suggest that: knowledge structure is important to understand how different parts of the task relate to each other, which is important for fast-paced, real-time, and complex tasks; knowledge content is important for more asynchronous tasks that require integration of knowledge from multiple task domains; and that effective knowledge structures promote the acquisition of knowledge content (Cannon-Bowers et al., 2001). Similarly, we have not discussed an attribute about knowledge depth. While internalization, differentiation, newness and accuracy are important attributes of the dimensions proposed, the resulting knowledge can be very superficial and vague or very deep and detailed. Nevertheless, we decided to focus on important dimensions that capture the essence of team knowledge, while avoiding over-complicating the construct at this stage.

REFERENCES

- Adams, M., Tenney, Y., & Pew, R. 1995. Situation awareness and the cognitive management of complex systems. *Human Factors*, 37(1), 85-104.
- Alavi, M., & Leidner, D. E. 2001. Knowledge management and knowledge management systems: Conceptual foundations and research issues. *MIS Quarterly*, 25(1): 107-136.
- Anand, V., Clark, M. A., & Zellmer-Bruhn, M. 2003. Team knowledge structures: Matching task to information environment. *Journal of Managerial Issues*, 5(1): 15-31.
- Anand, V., Glick, W. H., & Manz, C. C. 2002. Thriving on the knowledge of outsiders: Tapping organizational social capital. *Academy of Management Executive*, 16(1): 87-101.
- Anand, V., Manz, C. C., & Glick, W. H. 1998. An organizational memory approach to information management. *Academy of Management Review*, 23, 796-809.
- Austin, J. R. 2003. Transactive memory in organizational groups: The effects of content, consensus, specialization, and accuracy on group performance. *Journal of Applied Psychology*, 88, 866-878.
- Axelrod, R. (Ed.). 1976. *Structure of decision: The cognitive maps of political elites*. Princeton, NJ: Princeton University Press.
- Bantel, K. A. 1993. Top team, environment, and performance effects on strategic planning formality. *Group and Organization Management*, 18, 436-458.
- Baumann, M. R., & Bonner, B. L. 2004. The effects of variability and expectations on utilization of member expertise and group performance. *Organizational Behavior and Human Decision Processes*, 93, 89-101.
- Berman, S. L., Down, J., & Hill, C. W. L. 2002. Tacit knowledge as a source of competitive advantage in the national basketball association. *Academy of Management Journal*, 45, 13-31.
- Boje, D. M. 1991. The storytelling organization: A study of story performance. *Administrative Science Quarterly*, 36, 106-126.
- Boland, R., Singh, J., Salipante, P., Aram, J. D., Fay, S., & Kanawattanachai, P. 2001. Knowledge representations and knowledge transfer. *Academy of Management Journal*, 44, 393-417.
- Boyer, D. G., Handel, M. J., & Herbsleb, J. D. 1998. Virtual community presence awareness. *ACM SIGGROUIP Bulletin*, 19(3): 11-14.
- Cannon-Bowers, J. A., & Salas, E. 2001. Reflections on shared cognition. *Journal of Organizational Behavior*, 22, 195-202.
- Cannon-Bowers, J. A., Salas, E., & Converse, S. 1993. Shared mental models in expert team decision-making. In J. Castellan (Ed.), *Individual and group decision-making: Current issues*: 221-246: LEA Publishers.
- Carley, K. 1986. An approach for relating social structure to cognitive structure. *Journal of Mathematical Sociology*, 12, 137-189.
- Carley, K. 1997. Extracting team mental models through textual analysis. *Journal of Organizational Behavior*, 18, 533-558.

Chen, L., & Gaines, B. 1997. A cyber-organism model for awareness in collaborative communities on the internet. *International Journal of Intelligent Systems*, 12(1), 31-56.

Clark, H., & Carlson, T. 1982. Speech acts and hearers' beliefs. In N. V. Smith (Ed.), *Mutual knowledge*: 1-36. New York, NY: Academic Press.

Clark, H., & Marshal, C. 1981. Definite reference and mutual knowledge. In A. K. Joshi, I. Sag, & B. Webber (Eds.), *Elements of discourse and understanding*: 10-63. New York, NY: Cambridge University Press.

Clark, M. A., Anand, V., & Roberson, L. 2000. Resolving meaning: Interpretation in diverse decision-making groups. *Group Dynamics: Theory, Research, & Practice*, 4, 211-221.

Cohen, M. D. 1996. Individual learning and organizational routine: Emerging connections. In M. D. Cohen, & L. S. Sproull (Eds.), *Organizational learning*: 188-194. Thousand Oaks, CA: Sage.

Cohen, S. G., & Bailey, D. E. 1997. What makes teams work: Group effectiveness research from the shop floor to the executive suite. *Journal of Management*, 23: 239-290.

Cooke, N. J., Kiekel, P. A., Salas, E., Stout, R. J., Bowers, C., & Cannon-Bowers, J. A. 2003. Measuring team knowledge: A window to the cognitive underpinnings of team performance. *Group Dynamics: Theory, Research and Practice*, 7, 179-199.

Cooke, N. J., Salas, E., Cannon-Bowers, J. A., & Stout, R. J. 2000. Measuring team knowledge. *Human Factors*, 42(1): 151-173.

Corman, S. R., Kuhn, T., McPhee, R. D., & Dooley, K. J. 2002. Studying complex discursive systems: Centering resonance analysis of communication. *Human Communication Research*, 28, 157-206.

Cramton, C. D. 2001. The mutual knowledge problem and its consequences for dispersed collaboration. *Organization Science*, 12, 346-371.

Crowston, K., & Kammerer, E. E. 1998. Coordination and collective mind in software requirements development. *IBM Systems Journal*, 37(2): 227-245.

Curtis, B., Krasner, H., & Iscoe, N. 1988. A field study of the software design process for large systems. *Communications of the ACM*, 31(11): 1268-1286.

Daniels, K., Johnson, G., & Chernatony, L. d. 1994. Differences in managerial cognition of competition. *British Journal of Management*, 5, 21-29.

Davenport, T. H., De Long, D. W., Beers, M. C. (1998). Successful knowledge management projects. *Sloan Management Review*, 39(2): 43-57.

Doty, D. H., & Glick, W. H. 1994. Typologies as a unique form of theory building: Towards improved understanding and modeling. *Academy of Management Review*, 19, 230-251.

Dourish, P., & Bly, S. 1992. *Portholes: Supporting awareness in a distributed work group*. Paper presented at the ACM Conference on Human Factors in Computing Systems (INTERCHI).

Droege, S. B., & Hoobler, J. M. 2003. Employee turnover and tacit knowledge diffusion: A network perspective. *Journal of Managerial Issues*, 15(1): 50-64.

Ellis, A. P., Hollenbeck, J. R., Ilgen, D. R., Porter, C. O. L. H., West, B. J., & Moon, H. 2003. Team learning: Collectively connecting the dots. *Journal of Applied Psychology*, 88, 821-835.

Endsley, M. 1995. Toward a theory of situation awareness in dynamic systems. *Human Factors*, 37(1).

Espinosa, J. A. 2001. *Shared mental models: Accuracy and visual representation*. Paper presented at the Americas Conference in Information Systems, Boston MA.

Espinosa, J. A., Cadiz, J., Rico-Gutierrez, L., Kraut, R. E., Scherlis, W., & Lautenbacher, G. 2000. *Coming to the wrong decision quickly: Why awareness tools must be matched with appropriate tasks*. Paper presented at the Computer-Human Interaction (CHI), Association of the Computer Machinery's (ACM), The Hague, The Netherlands.

Espinosa, J. A., & Carley, K. 2001. *Measuring team mental models*. Paper presented at the Academy of Management Conference, Washington D.C.

Espinosa, J. A., Carley, K., Kraut, R. E., Lerch, F. J., & Fussell, S. 2002a. *The effect of task knowledge similarity and distribution on asynchronous team coordination and performance: Empirical evidence from decision teams*. Paper presented at the Information Systems Cognitive Research Exchange (IS CoRE) workshop, Barcelona, Spain.

Espinosa, J. A., Kraut, R. E., Slaughter, S. A., Lerch, F. J., Herbsleb, J. D., & Mockus, A. 2002b. *Shared mental models, familiarity, and coordination: A multi-method study of distributed software teams*. Paper presented at the International Conference in Information Systems, Barcelona, Spain.

Faraj, S., & Sproull, L. 2000. Coordinating expertise in software development teams. *Management Science*, 46, 1554-1568.

Fussell, S., & Krauss, R. 1992. Coordination of knowledge in communication: Effects of speakers' assumptions about what other know. *Journal of Personality and Social Psychology*, 62, 378-391.

Gersick, C. J., & Hackman, J. R. 1990. Habitual routines in task-performing groups. *Organizational Behavior and Human Decision Processes*, 47: 65-97.

Guerin, B. (1992). Behavior analysis and the social construction of knowledge. *American Psychologist*, 47, 1423-1432.

Gutwin, C., & Greenberg, S. 1999. The effects of workspace awareness support on the usability of real-time distributed groupware. *ACM Transactions on Computer-Human Interaction (ToCHI)*, 6(3): 243-281.

Gutwin, C., & Greenberg, S. 2004. The importance of awareness for team cognition in distributed collaboration. In E. Salas, & S. M. Fiore (Eds.), *Team cognition: Understanding the factors that drive process and performance*: 177-201, ch. 179, ISBN 171-59147-59103-59146. Washington, D.C.: American Psychological Association.

Hammer, M., Leonard, D., & Davenport, T. (2004). Why don't we know more about knowledge? *Sloan Management Review*, 45(4): 13-18.

Hansen, M. T. 1999. The search-transfer problem: The role of weak ties in sharing

knowledge across organization subunits. *Administrative Science Quarterly*, 44, 82-121.

Helmreich, R. L. 1997. Managing human error in aviation, *Scientific American*, 277: 40.

Hinsz, V. B., Tindale, R. S., & Vollrath, D. A. 1997. The emerging conceptualization of groups as information processors. *Psychological Bulletin*, 121, 43-64.

Jackson, S. E., Brett, J. F., Sessa, V. I., Cooper, D. M., Julin, J. A., & Peyronnin, K. 1991. Some differences make a difference: Individual dissimilarity and group heterogeneity as correlates of recruitment, promotions, and turnover. *Journal of Applied Psychology*, 76, 675-689.

James, L. R., Demaree, R. G., & Wolf, G. 1984. Estimating within-group interrater reliability with and without response bias. *Journal of Applied Psychology*, 69, 85-98.

Kelly, G. A. 1955. *The psychology of personal constructs*. New York, NY: Norton.

Klimoski, R. J., & Mohammed, S. 1994. Team mental model: Construct or metaphor. *Journal of Management*, 20, 403-437.

Kogut, B., & Zander, U. 1997. What firms do? Coordination, identity and learning. *Organization Science*, 7(5).

Kraiger, K., & Wenzel, L. 1997. Conceptual development and empirical evaluation of measures of shared mental models as indicators of team effectiveness. In M. Brannick, E. Salas, & C. Prince (Eds.), *Team performance assessment and measurement*: 63-84: LEA Publishers.

Krauss, R., & Fussell, S. 1990. Mutual knowledge and communicative effectiveness. In J. Galegher, R. E. Kraut, & C. Egidio (Eds.), *Intellectual teamwork: Social and technological foundations of cooperative work*: 111-146. Hillsdale, NJ: Lawrence Erlbaum.

Kuhn, T., & Corman, S. R. 2003. The emergence of homogeneity and heterogeneity of knowledge structures during a planned organizational change. *Communication Monographs*, 70(3): 198-229.

Lau, D., & Murnighan, J. K. 1998. Demographic diversity and faultlines: The compositional dynamics of organizational groups. *Academy of Management Review*, 23, 325-340.

Lawrence, P. R., & Lorsch, J. W. 1967. *Organization and environment*. Boston, MA: Harvard University Press.

Leibiskind, J. R., Oliver, A. L., Zucker, L., & Brewe, M. 1996. Social networks, learning and flexibility: Sourcing scientific knowledge in new bio-technology firms. *Organization Science*, 7, 428-443.

Levesque, L. L., Wilson, J. M., & Wholey, D. R. 2001. Cognitive divergence and shared mental models in software development project teams. *Journal of Organizational Behavior*, 22, 135-144.

Lewis, K. 2003. Measuring transactive memory systems in the field: Scale development and validation. *Journal of Applied Psychology*, 88, 587-604.

Libby, R., Trotman, K. T., & Zimmer, I. 1997. Member variation, recognition of expertise, and group performance. *Journal of Applied Psychology*, 72, 81-87.

Mathieu, J., Goodwin, G. F., Heffner, T. S., Salas, E., & Cannon-Bowers, J. A. 2000. The

influence of shared mental models on team process and performance. *Journal of Applied Psychology*, 85, 273-283.

McFadyyn, M. A., & A. A. Cannella, j. in press. Social capital and knowledge creation: Diminishing returns of the number and strength of exchange relationships. *Academy of Management Journal*, 47.

Murray, A. I. 1989. Top management team heterogeneity and firm performance. *Strategic Management Journal*, 10, 125-141.

Nelson, K. M., & Coopridge, J. G. 1996. The contribution of shared knowledge to is group performance. *MIS Quarterly*, 20(4): 409-432.

Nonaka, I. 1994. A dynamic theory of organizational knowledge creation. *Organization Science*, 5, 14-37.

Nonaka, I., & Takeuchi, H. 1995. *The knowledge creating company*. New York, NY: Oxford University Press.

Pelled, L., Eisenhardt, K., & Xin, K. 1999. Exploring the black box: An analysis of work group diversity, conflict, and performance. *Administrative Science Quarterly*, 44, 1-28.

Polyani, M. 1958. *Personal knowledge*. Chicago, IL: The University of Chicago Press.

Polyani, D.M. 1966. ***The Tacit Dimension***. London: Routledge and Kegan Paul.

Quinn, J. B. 1993. *Intelligent enterprise*. New York, NY: Free Press.

Reagans, R., & Zuckerman, E. W. 2001. Networks, diversity, and performance: The social capital of corporate R&D units. *Organization Science*, 12, 502-517.

Rentsch, J. R., & Hall, R. J. 1994. Members of great teams think alike: A model of the effectiveness and schema similarity among team members. *Advances in Interdisciplinary Studies of Work Teams*, 1: 223-261.

Rentsch, J. R., & Klimoski, R. J. 2001. Why do great minds think alike? Antecedents of team member schema agreement. *Journal of Organizational Behavior*, 22, 107-120.

Salas, E., Prince, C., Baker, D. P., & Shrestha, L. 1995. Situation awareness in team performance: Implications for measurement and training. *Human Factors*, 37(1): 123-136.

Schultz, M. (2001). The uncertain relevance of newness: Organizational learning and knowledge flows. *Academy of Management Journal*, 44, 661-681.

Smith, J. 1994. *Collective intelligence model in computer-based collaboration*. New Jersey: Lawrence Erlbaum Associates.

Sparrowe, R. T., Liden, R. C., Wayne, S. J., & Kraimer, M. L. 2001. Social networks and the performance of individuals and groups. *Academy of Management Journal*, 44(2): 316-325.

Stasser, G., Stewart, D. D., & Wittenbaum, G. W. 1995. Expert roles and information exchange during discussion: The importance of knowing who knows what. *Journal of Experimental Social Psychology*, 31: 244-265.

Stasser, G., & Titus, W. 1985. Pooling of unshared information in group decision making: Biased information sampling during discussion. *Journal of Personality and Social Psychology*, 48, 1467-1478.

Steinfeld, C., Jang, C.-Y., & Pfaff, B. 1999. *Supporting virtual team collaboration: The teamscope system*. Paper presented at the Group '99: International ACM SIGGROUP Conference on Supporting Group Work, Phoenix, AZ.

Stout, J. A., Cannon-Bowers, J., & Salas, E. 1999. Planning, shared mental models, and coordinated performance: An empirical link is established. *Human Factors*, 41: 61-71.

Tajfel, H., & Turner, J. C. 1985. The social identity theory of intergroup behavior. In S. Worchel, & W. G. Austin (Eds.), *The social psychology of intergroup relations*: 7-24. Chicago, IL: Nelson-Hall.

Tsai, W. 2002. Social structure of "coopetition" within a multiunit organization: Coordination, competition, and intraorganizational knowledge sharing. *Organization Science*, 13, 179-190.

Walz, D. B., Elam, J. J., & Curtis, B. 1993. Inside a software design team: Knowledge acquisition, sharing, and integration. *Communications of the ACM*, 36(10): 63-77.

Wegner, D. 1986. Transactive memory: A contemporary analysis of the group mind. In B. Mullen, & G. Goethals (Eds.), *Theories of group behavior*: Springer-Verlag.

Wegner, D. 1995. A computer network model of human transactive memory. *Social Cognition*, 13, 319-339.

Weick, K. 1990. The vulnerable system: An analysis of the tenerife air disaster. *Journal of Management*, 16, 571-593.

Weick, K. 1993. The collapse of sensemaking in organizations: The Mann Gulch disaster. *Administrative Science Quarterly*, 38, 628-652.

Weick, K., & Roberts, K. 1993. Collective mind in organizations: Heedful interrelating on flight decks. *Administrative Science Quarterly*, 38, 357-381.

Weick, K. E., & Bougon, M. G. 1986. Organizations as cognitive maps: Charting ways to success and failure. In H. P. Sims, & D. A. Gioia (Eds.), *The thinking organization*. San Francisco, CA: Jossey-Bass.

Wellens, R. 1993. Group situation awareness and distributed decision-making: From military to civilian applications. In J. Castellan (Ed.), *Individual and group decision-making: Current issues*: 267: LEA Publishers.

Wittenbaum, G. M., & Stasser, G. 1996. Management of information in small groups. In J. L. Nye, & A. M. Brower (Eds.), *What's social about social cognition?*, 3: Sage Publications.

Zenger, T. R., & Lawrence, B. S. 1989. Organizational demography: The differential effects of age and tenure distributions on technical communication. *Academy of Management Journal*, 32, 353-376.

Table 1

Individual/Shared vs. Long-Term/Dynamic Knowledge

	Long-Term	Dynamic
Individual	Individual Long-Term Knowledge	Individual Situation Awareness
Shared	Shared Long-Term Knowledge	Team Awareness

Table 2

Taskwork/Teamwork and Long-Term/Dynamic Knowledge

	Long-Term	Dynamic
Taskwork	Long-Term Taskwork Knowledge	Team Awareness of Taskwork
Teamwork	Long-Term Teamwork Knowledge	Team Awareness of Teamwork

Table 3

Explicit/Tacit and Taskwork/Teamwork Knowledge

	Explicit	Tacit
Taskwork	Declarative Task Knowledge	Procedural Knowledge / Ability
Teamwork	Team Process Knowledge	Team Process Ability